London Bicycles

Data Engineering Insight, Risk & Surprise Findings Report

GCP Project: **m2-dataset-study** | Dataset: **bigquery-public-data.london_bicycles** | Generated: **19 May 2026**

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1. Executive Summary

This report analyses the Transport for London (TfL) Santander Cycles public dataset on Google BigQuery, spanning **83.4 million rental records** from January 2015 to January 2023 across **~800 docking stations**. It surfaces actionable insights, proposes a production-grade data warehouse architecture, identifies key risks, and highlights four genuinely unexpected findings uncovered during the study.

Central business problem: *TfL operates one fleet serving two completely different markets — commuters and leisure/tourist riders — without optimising fleet rebalancing for either segment.* This creates predictable, preventable supply failures every day.

Metric	Value
Total rental records	83,434,866
Date range	4 Jan 2015 – 15 Jan 2023
Active stations (peak)	~800
Unique bikes ever tracked	~24,700
Avg trip duration (all years)	21.9 minutes
Peak month (Jul 2022)	1,303,376 rentals
Trough month (Jan 2022)	741,875 rentals
Missing end-station records	561,495 (0.67%)

2. Key Insights from the Data

2.1 Two Markets, One Fleet

Usage patterns differ fundamentally by day type, exposing two distinct user segments sharing the same docking infrastructure:

Hour	Weekday Rentals	Weekend Rentals	Ratio W/E
08:00	7,156,970	412,928	17×
12:00–14:00	~5.7M	~5.8M	~1×
17:00–18:00	13,221,451	~3.1M	4×

- **Weekday:** bimodal spikes at 08:00 and 17–18:00 → commuters using bikes as last-mile transport.
- **Weekend:** flat plateau 10:00–17:00 → leisure riders and tourists with no time pressure.

2.2 Top Routes Are All Leisure Circuits

The 10 highest-volume routes are closed loops inside Hyde Park and Kensington Gardens. No commuter route appears in the top 10.

Route (Start → End)	Trips	Avg Duration
Hyde Park Corner → Albert Gate	33,122	42.2 min
Hyde Park Corner → Triangle Car Park	31,747	29.7 min
Black Lion Gate → Palace Gate	30,908	23.5 min
Albert Gate → Hyde Park Corner	29,861	42.5 min
Aquatic Centre → Podium (Olympic Park)	23,501	42.9 min

2.3 COVID Signature in Trip Duration (2020)

Year	Rentals	Avg Duration (min)	Unique Bikes
2018	10,441,884	19.9	12,934
2019	10,310,063	19.5	12,943
2020	**10,287,040**	**26.2** ↑	**14,743**
2021	10,821,289	21.9	14,962
2022	11,027,519	21.3	24,715

The 34% duration spike in 2020 is consistent with lockdown-era behaviour: commuter trips disappeared and people rode bikes purely for outdoor exercise. The pipeline should flag month-over-month duration deviations > 15%.

2.4 Seasonal Demand — 3× Summer vs Winter (2022)

Month	Rentals	Round Trips	Trips > 1hr
January	741,875	24,749	15,899
April	1,022,467	43,469	41,662
July	1,303,376	45,017	51,014
September	609,978	20,453	14,235

Month	Rentals	Round Trips	Trips > 1hr
December	421,860	13,449	8,602

September 2022 is anomalously low (609k vs 835k in October), aligning with area closures after Queen Elizabeth II's death (8 September 2022). Maintain a seed table of external events to contextualise such drops.

2.5 E-Bike vs Classic — Different Behaviour Profiles

Bike Model	Rentals	Avg Duration	Unique Bikes	Coverage
NULL/Untagged	81,048,143	~21.9 min	~20,779	97.1% of all records
CLASSIC	2,254,688	23.5 min	12,284	2022–2023 only
PBSC_EBIKE	132,035	18.9 min	492	2022–2023 only

E-bikes show shorter durations (18.9 min) suggesting commuter use. 97% of records lack model tags — scope e-bike analysis to 2022+ only.

3. Surprise Findings

The following four findings were not part of the original hypothesis. They emerged directly from querying the data and represent unexpected behaviours with significant implications for the pipeline design, data quality strategy, and business recommendations.

3.1 Maintenance Trips Hidden Inside the Rental Record

★ The Surprise	Three stations named 'Mechanical Workshop Clapham', 'Mechanical Workshop Penton', and 'Electrical Workshop PS' appear as end destinations for real rental records. Bikes sent to these workshops show average 'trip' durations of 7,889–10,707 minutes (131–178 hours). The longest single rental record in the entire dataset is 65.3 days — a bike tagged as rented from 'Westbourne Park Road' travelling to 'West Smithfield Rotunda'. It never made a customer trip. It was in maintenance custody.

Workshop Destination	Trips Recorded	Unique Bikes	Avg 'Duration'
Mechanical Workshop Clapham	777	757	9,076 min (151 hrs)
Mechanical Workshop Penton	724	706	7,889 min (131 hrs)
Electrical Workshop PS	2	2	10,707 min (178 hrs)

Why it matters: These records contaminate duration averages, inflate the 'trips > 24 hours' anomaly count, and skew any route analysis that includes workshop destinations. Without filtering, maintenance logistics appear as legitimate customer journeys.

What to do: In stg_cycle_hire, add a boolean flag *is_maintenance_trip* for any record where end_station_name contains 'Workshop'. Route these to a separate mart_maintenance_trips. This table becomes an asset — it reveals which bikes were serviced, how frequently, and the gap between last rental and workshop arrival.

3.2 Saturday Night Is a Hidden Third Market

★ The Surprise	Midnight on Sunday (Saturday night / early Sunday morning) generates 210,272 rentals across the full dataset — 2.3× more than Tuesday midnight (73,963) and nearly matching the Friday midnight volume. Late-night trips (00:00–04:00) are also consistently longer than average (31–38 min vs the fleet mean of 21.9 min), suggesting riders are cycling home from entertainment districts, not commuting.

Day	Midnight Rentals	2am Rentals	Avg Night Duration
Sunday (Sat night)	**210,272**	**110,872**	**32.8 min**
Saturday (Fri night)	196,682	99,461	32.7 min
Friday	116,406	41,649	31.6 min
Thursday	92,635	32,598	30.9 min
Monday	91,517	29,995	31.4 min
Tuesday	73,963	25,163	28.6 min

Why it matters: The current commuter/leisure segmentation model has a blind spot: it misclassifies Saturday-night trips as 'leisure' when they are actually a distinct nightlife segment with different station needs (entertainment hubs: Soho, Shoreditch, Waterloo) and longer trip distances.

What to do: Add a third trip_segment value — *nightlife* — defined as 00:00–04:00 on Friday and Saturday nights. Build a mart_nightlife_demand that ranks the top departure stations for this window. This is an untapped rebalancing optimisation: pre-position bikes near nightlife hubs before midnight.

3.3 Waterloo Bleeds Bikes; Borough Absorbs Them — Permanently

★ The Surprise	Across 8 years of data, Waterloo Station 2 shows a net outflow of +69,147 bikes (16.2% imbalance) — far more bikes leave than arrive. The opposite extreme is Hop Exchange (Borough): a net inflow of –132,544 bikes (–15.0% imbalance). These are not temporary fluctuations. They are structural, permanent patterns driven by the urban geography: train passengers grab bikes at Waterloo and cycle to offices near London Bridge. The bikes never come back.

Station	Departures	Arrivals	Net Flow	Imbalance
Hop Exchange, The Borough	376,225	508,769	–132,544	–15.0% (sink)
Holborn Circus, Holborn	222,739	304,269	–81,530	–15.5% (sink)
Waterloo Station 2	**247,486**	**178,339**	**+69,147**	**+16.2% (source)**
Eagle Wharf Road, Hoxton	215,689	174,205	+41,484	+10.6% (source)
Cloudesley Road, Angel	89,758	54,506	+35,252	+24.4% (source)

Why it matters: No rebalancing algorithm can fix a structural imbalance — it requires physical van redistribution on a daily, predictable schedule. The data proves this is not an edge case: Waterloo has been a net source for 8 years. The mart_station_demand model should flag *permanently imbalanced* stations (imbalance > 10% over rolling 90 days) separately from stations with temporary demand spikes.

What to do: Add an is_structural_imbalance flag to dim_station, calculated from 90-day rolling net flow. Surface this in the executive dashboard as 'stations requiring daily van redistribution' — a direct, quantifiable operational cost that the data pipeline makes visible for the first time.

3.4 Super-Bikes: 8,197 Rentals on One Frame Over 7.7 Years

★ The Surprise	Bike #12942 has been rented 8,197 times between January 2015 and September 2022 — roughly 3 rentals per day for 7.7 years — accumulating 2,810 hours (117 days) of actual riding time. The top 10 hardest-working bikes all show 7,600–8,200 rentals with active lifespans of 2,555–2,805 days. None of these bikes appear in workshop destination records, meaning no maintenance event is recorded for them.

Bike ID	Total Rentals	Total Hours	Avg Trip (min)	Active Lifespan
12942	8,197	2,810 hrs	20.6	2,800 days (7.7 yrs)
11077	8,063	2,701 hrs	20.1	2,805 days
10601	7,970	2,749 hrs	20.7	2,719 days

Bike ID	Total Rentals	Total Hours	Avg Trip (min)	Active Lifespan
12779	7,941	2,760 hrs	20.9	2,805 days
12926	7,716	2,471 hrs	19.2	2,791 days

Why it matters: The data does not record any workshop visit for these bikes. Two interpretations: (a) they genuinely never went to maintenance — a serious safety and mechanical concern for bikes with 7+ years of continuous use; or (b) workshop records exist but the bike_id was not captured, exposing a data linkage gap between the rental system and the maintenance system.

What to do: Build mart_bike_utilisation to compute cumulative rentals and total hours per bike_id. Define a retirement threshold (e.g. > 5,000 rentals or > 2,000 hours) and surface bikes that exceed it with no associated workshop record. This analysis alone justifies a pipeline delivering operational safety value — a strong executive talking point beyond analytics.

4. Proposed Data Pipeline Architecture

Modern ELT pattern entirely within BigQuery (EU region). No data copied from the public dataset — all transformations create materialised views or tables inside m2-dataset-study.

Layer	Tool	Output
Source	bigquery-public-data (read-only)	Raw cycle_hire, cycle_stations
Staging	dbt — stg_cycle_hire, stg_stations	Cleaned, typed, maintenance-flagged
Dimensions	dbt — dim_date, dim_station, dim_bike	Conformed dimension tables
Fact	dbt — fact_rentals	83M row central fact (excl. maintenance)
Marts	dbt — 6 mart models	Station demand, routes, nightlife, bikes
Analysis	Python / pandas / SQLAlchemy	Jupyter notebooks & charts
Orchestration	GitHub Actions (cron) — optional	Scheduled dbt runs & quality tests
Quality	dbt tests + Great Expectations	Structural + statistical checks

4.1 Star Schema Design

```
Fact table: fact_rentals
rental_id, bike_id (FK), start_station_id (FK), end_station_id (FK),
date_id (FK), duration_sec, duration_min, is_round_trip,
trip_segment, -- commute_am | commute_pm | leisure | nightlife | night
is_maintenance_trip, -- NEW: flags workshop-destination records
Dimension tables:
dim_station (id, name, lat, lon, docks_count, is_temporary,
is_structural_imbalance) -- NEW flag from 90-day net flow
dim_bike (bike_id, bike_model, model_category, total_rentals,
total_hours, exceeds_retirement_threshold) -- NEW
dim_date (date_id, date, year, month, day_of_week,
season, is_weekend, is_uk_bank_holiday, is_disrupted_period)
```

5. Risk Register & Mitigations

Each risk is rated Likelihood (L) and Impact (I) on a 1–3 scale, derived from direct evidence in the dataset.

R1: Station Metadata Gap — 52% of hire stations have no dim_station record	Likelihood: **High** | Impact: **High**
Finding	cycle_hire references 1,625 unique station IDs; cycle_stations contains only 785. 840 stations (52%) have no latitude, longitude, or dock count — making geo-spatial and capacity analyses incomplete for over half the network.
Mitigation	Enrich dim_station from the TfL Unified API (bikepoint endpoint). Schedule a weekly dbt seed refresh. Flag unresolved stations as 'unknown_station'; never silently NULL-fill with incorrect data.

R2: bike_model NULL for 97% of Records — Limits Fleet Segmentation	Likelihood: High | Impact: Medium
Finding	Only 2.86% of records carry a non-null bike_model. E-bike vs Classic analysis is valid only for 2022–2023. Applying model-based insights to the full 2015–2022 history is misleading.
Mitigation	Scope all bike_model analysis explicitly to 2022+. Map NULL/empty → 'Unknown' in dim_bike. Document the coverage gap prominently in mart YAML descriptions.

R3: Duration Outliers Inflate KPIs — Including Maintenance Masquerade	Likelihood: High | Impact: Medium
Finding	47,133 trips (0.056%) exceed 24 hours. Workshop 'rentals' average 131–178 hours. Unfiltered, these inflate duration averages and contaminate route analysis. The longest single record is 65.3 days.
Mitigation	Flag is_maintenance_trip in staging (end_station_name LIKE '%Workshop%'). Apply a 24-hr duration cap for customer analytics. Route both anomaly types to separate mart tables — they have different operational value.

R4: Missing End Stations — 561,495 Records Unresolvable	Likelihood: Medium | Impact: Medium
Finding	0.67% of trips have NULL end_station_id. Likely causes: bike returned outside a dock, system error, or theft. Cannot be used for route or flow-balance analysis.
Mitigation	Exclude from route and flow-balance marts. Include in volume KPIs. Add a dbt test with warn threshold 0.7%, error at 1%.

R5: EU Region Lock — All Resources Must Be in EU	Likelihood: Medium | Impact: High
Finding	london_bicycles is stored in BigQuery EU multi-region. Any dataset created in m2-dataset-study outside EU will cause cross-region query failures.
Mitigation	Set location: EU in dbt profiles.yml. Create all datasets with --location=EU. Add a CI check validating dataset location before any dbt run.

R6: Read-Only Source — No Backfill or Correction Possible	Likelihood: Medium | Impact: Medium
Finding	bigquery-public-data is Google-managed. The project cannot correct errors or control freshness. Data ends January 2023 with no update guarantee.
Mitigation	Design dbt models as idempotent and incremental. Document the cutoff date. For real-time needs, supplement with the TfL live feed via Cloud Pub/Sub.

R7: External Event Contamination — Anomalies Silently Distort KPIs	Likelihood: Low | Impact: Medium
Finding	September 2022 shows 609k rentals — 27% below October — due to area closures after the Queen's death. Without context, automated reports flag this as a pipeline error.
Mitigation	Maintain a dbt seed external_events.csv with dated disruptions. Join to mart_monthly_trends and add is_disrupted_period. Alert consumers to exclude flagged months from baseline calculations.

R8: Structural Station Imbalance Cannot Be Solved by Algorithms	Likelihood: High | Impact: High
Finding	Waterloo has a permanent +69k net bike outflow; Hop Exchange has –132k inflow over 8 years. No rebalancing model can fix structural geographic flow — it requires physical van redistribution on a daily schedule.
Mitigation	Add is_structural_imbalance to dim_station (imbalance > 10% over 90-day roll). Surface in executive dashboard as 'stations requiring daily van redistribution'. Separate from stations with temporary demand spikes in all mart models.

6. ELT Transformation Plan (dbt)

dbt Model	Type	Key Logic
stg_cycle_hire	View	Cast types; filter duration ≤ 0; flag is_maintenance_trip; flag missing end_station
stg_stations	View	Join cycle_stations + TfL API seed; flag unresolved stations
dim_date	Table	Date spine 2015–2023; season, is_weekend, is_uk_bank_holiday, is_disrupted_period
dim_station	Table	Deduplicate; geo fields; is_temporary; is_structural_imbalance (90-day net flow)
dim_bike	Table	Map NULL model → Unknown; cumulative rentals/hours; exceeds_retirement_threshold
fact_rentals	Incremental	All dims joined; duration_min, is_round_trip, trip_segment (incl. nightlife)
mart_station_demand	Table	Hourly net flow per station; flag structural vs temporary imbalance
mart_nightlife_demand	Table	NEW: Fri/Sat 00:00–04:00 top departure stations
mart_bike_utilisation	Table	NEW: Cumulative rentals & hours per bike; retirement flag
mart_monthly_trends	Table	Monthly KPIs; join external_events; is_disrupted_period
mart_route_volume	Table	Top routes; exclude maintenance + missing end stations

7. Data Quality Testing Plan

Test	Tool	Threshold	Action on Fail
fact_rentals.duration_min > 0	dbt singular	0 failures	Block pipeline
fact_rentals.rental_id unique + not null	dbt generic	0 failures	Block pipeline
end_station_id null rate < 1%	dbt test	Warn 0.7%, Error 1%	Alert only
is_maintenance_trip count plausible (< 2k)	dbt singular	Warn on breach	Investigate
Monthly rental count within 3σ of mean	Great Expectations	Warn on breach	Alert + flag mart
avg duration change < 30% MoM	Great Expectations	Warn on breach	Alert + investigate
No trips with start_date > end_date	dbt singular	0 failures	Block pipeline

8. Python Analysis — Recommended Notebook Structure

Notebook	Analysis	Business Question
01_demand_patterns.ipynb	Heatmap: hour × day_type rental counts	When should we rebalance?
02_station_flow.ipynb	Net flow per station; structural vs temporary	Which stations empty/overflow?
03_seasonality.ipynb	Monthly trend + 3σ anomaly detection	How should fleet size scale?
04_nightlife_segment.ipynb	NEW: Fri/Sat midnight top stations & routes	Where is the third market?
05_bike_utilisation.ipynb	NEW: Cumulative hours per bike; retirement list	Which bikes need replacing?

Connect via SQLAlchemy: *create_engine("bigquery://m2-dataset-study")*. Always query marts, never the raw 83M-row fact table directly.

9. Executive Presentation Outline (10 min)

Slide	Title	Key Message	Time
1	The Problem	One fleet, two markets — plus a hidden nightlife segment	1 min
2	The Evidence	Weekday vs Weekend demand shape; Saturday midnight spike	1.5 min
3	The Opportunity	3× seasonal swing; Waterloo/Borough structural imbalance	1 min
4	Surprise Findings	Maintenance masquerade; super-bikes; nightlife; imbalance	1.5 min
5	Our Architecture	ELT pipeline: BigQuery → dbt → Python → insights	1.5 min
6	Risks & Controls	Top risks (R1, R3, R5, R8) + mitigations	1 min
7	Recommendations	Segment-aware rebalancing + nightlife pre-positioning + bike retirement model	1 min
8	Q&A;	Ready to answer	5 min

ROI framing: Even a 1% improvement in bike availability during the 08:00–09:30 peak (~7.2M trips/year in that window) translates to ~72,000 additional rentals. At TfL's £2–3 average net revenue per rental, that is £144k–£216k incremental annual revenue from rebalancing optimisation alone. The nightlife segment (210k Saturday-midnight trips annually) represents a further 10–15% uplift opportunity if pre-positioned correctly.

10. Recommended Tech Stack & Justification

Component	Choice	Why
Cloud platform	GCP / BigQuery (EU)	Source data already here; zero egress cost; serverless scale
Transformation	dbt Core	SQL-native; version-controlled; built-in testing; free tier
Orchestration	GitHub Actions (cron)	Free; no infra; sufficient for batch cadence
Analysis	Python + pandas + BigQuery connector	Assignment requirement; Jupyter-friendly

Component	Choice	Why
Data quality	dbt tests + Great Expectations	dbt for structural; GE for statistical anomaly detection
Diagramming	Excalidraw	Free; clean architecture diagrams; SVG export
Documentation	dbt docs generate	Auto-generates data catalog from model YAML